Piatetski–Shapiro primes
Compact Lean blueprint

Manuscript of Hamed Mousavi; project checkpoint, 30 October 2026.

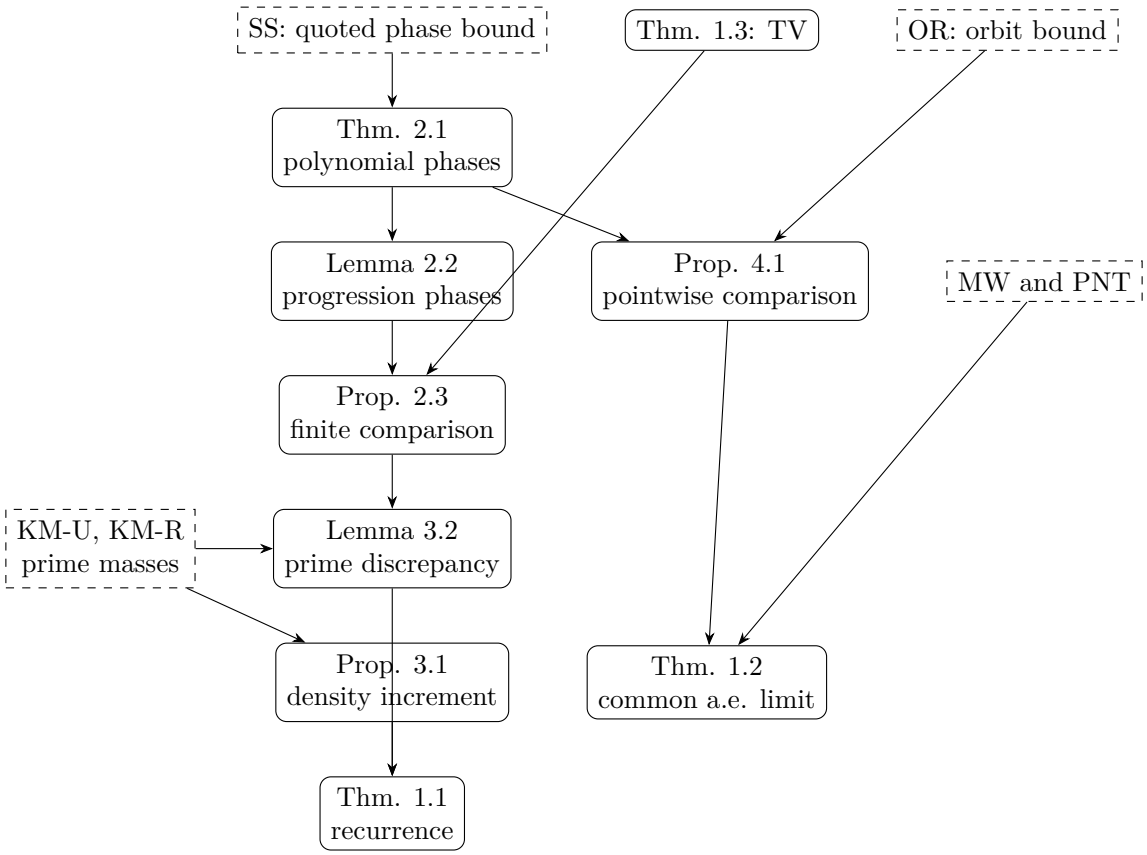
Reading and verification contract

This is the primary reading guide for Hamed Mousavi's *Polynomial configurations and pointwise averages along Piatetski–Shapiro primes*. Source: unchanged `Draft1.tex`, supplied 30 October 2026; compiled manuscript: 14 pages. No version or date is printed in the source. The source is preserved unchanged; its locally compiled PDF supplies the numbering. No independent supplied PDF was available.

Verified boundary. Theorem 1.2, Lemma 2.2 and Propositions 2.3 and 4.1 have checked conditional proofs. Theorem 1.1, Proposition 3.1 and Lemma 3.2 remain open. Theorems 1.3 and 2.1 remain external inputs; auxiliary proofs are recorded in the project evidence. A specification is not a theorem proof.

Four milestones organize the dependencies: exact objects and algebra; finite comparison; density increment and recurrence; pointwise comparison and unweighting. The longer companion explains every substantive step and remaining obligation.

Dependency graph



Dashed boxes are external inputs; solid boxes are manuscript nodes. Solid borders do not mean checked Lean proofs. The pointwise branch also needs prime-power removal, interpolation and unweighting; the recurrence branch needs exact affine transport and quantitative scale control. The web version generates the declaration DAG from the `uses/proves` metadata.

1 Main statements and common objects

Definition D.1 (The polynomial family and the two prime weights)**.** Fix $\ell \geq 1$ and $P_0 = 0$, $P_i \in \mathbb{Z}[y]$, with $P_i(0) = 0$ and

$$1 \leq \deg P_1 < \cdots < \deg P_\ell = d.$$

For $c > 1$, $\gamma = 1/c$, put

$$\mathcal{S}_c = \{\lfloor n^c \rfloor : n \in \mathbb{N}_{>0}\}, \quad \mathbb{P}_c = \mathcal{S}_c \cap \mathbb{P}, \quad \Lambda_\gamma(n) = \gamma^{-1} n^{1-\gamma} \Lambda(n) \mathbf{1}_{\mathcal{S}_c}(n), \quad D_\gamma = \Lambda_\gamma - \Lambda.$$

Weights are zero at nonpositive integer arguments. Set

$$\|w\|_{u^s[M]} = \sup_{\substack{\phi \in \mathbb{R}[y] \\ \deg \phi < s}} \left| M^{-1} \sum_{m=1}^{M} w(m) \mathrm{e}(-\phi(m)) \right|, \qquad \mathrm{e}(t) = \exp(2\pi i t).$$

Status: prime and polynomial objects implemented; the displayed u^s supremum is manuscript notation only.

Theorem 1.1 (Quantitative recurrence; manuscript Theorem 1.1)**.** *For every fixed family above there exist $c_* > 1$ and $\kappa > 0$ such that, for every $1 < c < c_*$, there is $N_0(\mathbf{P}, c)$ with the following property. If $N \geq N_0$ and $A \subset [N]$ satisfies*

$$|A| \geq N/(\log N)^\kappa,$$

then, for some $p \in \mathbb{P}_c$ and $x \in \mathbb{Z}$, all the elements

$$x,\ x + P_1(p-1), \ldots, x + P_\ell(p-1)$$

belong to A and are pairwise distinct.

Status: open; faithful specification exists.

Theorem 1.2 (Pointwise prime averages; manuscript Theorem 1.2)**.** *After decreasing $c_* > 1$ if needed, for every $1 < c < c_*$, every invertible probability measure-preserving system (X, ν, T) and every $f_1, \ldots, f_\ell \in L^\infty(X)$, the averages*

$$\frac{1}{\pi_c(N)} \sum_{\substack{p \leq N \\ p \in \mathbb{P}_c}} \prod_{i=1}^{\ell} f_i(T^{P_i(p-1)} x)$$

converge a.e. to the same limit as the analogous average over all primes, normalized by $\pi(N)$. Here $\pi_c(N) = |\mathbb{P}_c \cap [N]|$.

Status: checked conditional proof from SS, OR, PNT and MW; full uniform c_* specification.

Theorem 1.3 (Finite polynomial estimate; manuscript Theorem 1.3)**.** *For fixed integer polynomials $R_1, \ldots, R_\ell$ of distinct increasing degrees, largest degree d, and $C_0 \geq 1$, there are $K \geq 1$, $K \ll_d 1$, and a constant $C = C(C_0, \mathbf{R})$ such that, for all $M \geq 1$, 1-bounded F_i supported on $[-C_0 M^d, C_0 M^d]$ and 1-bounded θ on $[M]$,*

$$\left| M^{-d-1} \sum_{x \in \mathbb{Z}} \sum_{m=1}^{M} \theta(m) F_0(x) \prod_{i=1}^{\ell} F_i(x + R_i(m)) \right| \leq C\big(M^{-1} + \|\theta\|_{u^{d+1}[M]}\big)^{1/K}.$$

Status: external input TV, not a proved project theorem. The actual `FiniteVNInput` uses universal phase bounds and integer scales, with positive degree. Its source-to-interface bridge TV-B (supremum, sign and monotonicity) is explicitly unformalized. Its constant depends on the fixed family. Scaling $\theta = g/B$ gives the bound $CB(M^{-1} + B^{-1}\|g\|_u)^{1/K}$ when $|g| \leq B$ and $B > 0$. This scaling is an elementary deduction, not a strengthened external input.

2 Arithmetic and finite comparison

Theorem 2.1 (Polynomial-phase saving; manuscript Theorem 2.1)**.** *For every integer $r \geq 1$ there are $\varepsilon_r, \sigma_r > 0$ such that, whenever $1 - \varepsilon_r < \gamma < 1$,*

$$\sup_{\deg \phi \leq r} \left| \sum_{n=1}^{X} D_\gamma(n) \mathrm{e}(\phi(n)) \right| \ll_{r,\gamma} X^{1-\sigma_r}.$$

The implied constant is uniform over all real polynomial coefficients.

Status: external input SS: directly quoted polynomial-phase consequence.

Mathematical derivation and formalization boundary. Take $G = \mathbb{R}$, $\Gamma = \mathbb{Z}$, $G_0 = \cdots = G_r = \mathbb{R}$ and $G_j = \{0\}$ for $j > r$. Set $F(t + \mathbb{Z}) = \mathrm{e}(t)$. The filtered nilmanifold and Lipschitz bounds depend only on r. Expand $\phi(n) = \sum_{j=0}^{r} a_j \binom{n}{j}$. This is an adapted polynomial sequence, so SS gives the estimate with constants independent of the a_j. No quantitative Gowers-uniformity statement for D_γ is used. □

Lemma 2.2 (Restriction to progressions; manuscript Lemma 2.2)**.** *With the same admissible γ range and some positive saving σ_r, for all integers $Q, M \geq 1$,*

$$\sup_{\deg \psi \leq r} \left| \sum_{y=1}^{M} D_\gamma(Qy + 1) \mathrm{e}(\psi(y)) \right| \ll_{r,\gamma} (QM)^{1-\sigma_r}.$$

Status: checked conditional proof from `ShaoSunPhaseInput`; saving may be decreased.

Restriction argument. Write $L = QM$ and reindex by $n = Qy + 1$. Insert

$$\mathbf{1}_{n \equiv 1 \pmod{Q}} = Q^{-1} \sum_{a=0}^{Q-1} \mathrm{e}(a(n-1)/Q).$$

For each a the phase $\psi((n-1)/Q) + a(n-1)/Q$ has degree at most r, because $r \geq 1$. Including $n = 1$ causes no error since $D_\gamma(1) = 0$. Apply Theorem 2.1 at $L + 1$ to each character and average; the factor Q cancels exactly. □

Definition D.2 (Rescaling and the finite counting form)**.** For $P(0) = 0$ and integer $Q \geq 1$, define the integer polynomial

$$P_Q(y) = P(Qy)/Q = \sum_{j=1}^{\deg P} a_j Q^{j-1} y^j, \qquad P(y) = \sum_{j \geq 1} a_j y^j.$$

For finitely supported $f_i : \mathbb{Z} \to \mathbb{C}$ set

$$\mathcal{T}_{N,M,Q}(w; \mathbf{f}) = \frac{1}{NM} \sum_{x} \sum_{y=1}^{M} w(Qy + 1) f_0(x) \prod_{i=1}^{\ell} f_i(x + P_{i,Q}(y)).$$

Status: exact algebra checked; parent Proposition 2.3 conditionally checked.

Proposition 2.3 (Finite comparison; manuscript Proposition 2.3)**.** *Fix $C_0 \geq 1$, $0 < a < 1$ and the original family. There are $c_*(d) > 1$ and $\xi(d) > 0$ such that, for $1 < c < c_*$,*

$$Q \leq \exp((\log N)^a), \qquad M = \left\lfloor (N/Q^{d-1})^{1/d} \right\rfloor$$

imply, for sufficiently large M in terms of $c, \mathbf{P}, C_0, a$ and 1-bounded inputs supported on $[-C_0 N, C_0 N]$,

$$|\mathcal{T}_{N,M,Q}(D_\gamma; \mathbf{f})| \ll_{c,\mathbf{P},C_0} N^{-\xi}.$$

Status: checked conditional proof from SS and TV; TV-B remains external.

Fixed-family lift. Set $L = QM$. Lift f_i by $F_i(Qx) = f_i(x)$ and $F_i(z) = 0$ otherwise. Set

$$B = C_\gamma L^{1-\gamma}\log(2L), \qquad \Theta(n) = B^{-1}D_\gamma(n+1)\mathbf{1}_{Q|n}.$$

Then $|\Theta| \le 1$ and

$$\|\Theta\|_{u^{d+1}[L]} \ll B^{-1}L^{-\sigma_d}.$$

The lifted count uses the original fixed P_i and has normalization loss

$$\frac{L^{d+1}}{NM} \asymp Q^2.$$

TV therefore bounds the original count by

$$\ll BQ^2\big(L^{-1} + B^{-1}L^{-\sigma_d}\big)^{1/K}.$$

Choose $1-\gamma < \min(1,\sigma_d)/(8K)$; use $B \ll_\gamma L^{2(1-\gamma)}$ and absorb $Q^2 = L^{o(1)}$. The remaining negative power of $L = N^{1/d+o(1)}$ gives the claimed saving. □

3 Density increment and recurrence

Definition D.3 (Coefficient and modulus conditions)**.** Write $\mathcal{L}(s) = \operatorname{lcm}(1,\dots,s)$ and $P^+(1) = 1$. A positive-degree polynomial $R(y) = \sum_{j=1}^{e} a_j y^j$ has (C,Q)-coefficients if

$$a_e = a'_e Q^{e-1}, \quad 0 < |a'_e| \le C, \quad |a_j| \le C|a_e| \quad (1 \le j < e).$$

A single C bounding all coefficients of the original family works for every rescaling $P_{i,Q}$ and every positive Q.

Proposition 3.1 (Discrepancy gives an increment; manuscript Proposition 3.1)**.** *For fixed C, d, ℓ there are $C_1, K_1 \ge 2$, $0 < c_0 < 1/2$ and $c_1, c_2 > 0$ such that the following holds for sufficiently large N. Suppose*

$$(\log N)^{-1/C_1^2} < \delta \le 1, \quad \lceil K_1\delta^{-C_1}\rceil \le s \le (\log N)^{c_0},$$
$$Q \le \exp((\log N)^{2c_0}), \quad P^+(Q) \le s, \quad \mathcal{L}(s) \mid Q.$$

Let R_i have the specified distinct degrees and (C,Q)-coefficients, $R_0 = 0$, and put $M = \lfloor (N/Q^{d-1})^{1/d}\rfloor$. If $|A| = \delta N$ and the absolute prime-weighted discrepancy between $\mathbf{1}_A$ and $\delta\mathbf{1}_{[N]}$ is at least

$$c_1\delta^{\ell+1}\mathbb{E}_{y\le M}\Lambda(Qy+1),$$

then an atom $a + Q^b[N'] \subset [N]$, with $1 \le b \ll_d 1$, has density at least $(1+c_2)\delta$ and length

$$N' \ge Q^{-O_{d,\ell}(1)}\delta^{O_{d,\ell}(1)}M^r\exp(-d\sqrt{\log M}), \qquad r = \deg R_1.$$

Status: composite conditional deduction open; density algebra auxiliaries available.

The original local-factor deduction. KM-U gives $Me^{-\sqrt{\log M}} \le M_0 \le M$ and unweighted discrepancy at least $(c_1/4)\delta^{\ell+1}$. Apply KM-R with $\varepsilon = (c_1/16)\delta^{3(\ell+1)}$. Its small-scale alternative contradicts $\log M_0 \gg_d \log N$. Otherwise let F_i be the densities on its progression partitions. They preserve mass, and the approximation leaves discrepancy at least $(c_1/8)\delta^{\ell+1}$.

If every $F_i \le (1+c_2)\delta$, equality of masses gives

$$\sum_x |F_i(x) - \delta\mathbf{1}_{[N]}(x)| \le 2c_2\delta N.$$

Telescoping the products bounds the discrepancy by $2(\ell+1)c_2(1+c_2)^\ell\delta^{\ell+1}$, a contradiction when c_2 is small. Some atom therefore has the required density. KM-R's resolution and the bound on M_0 give the length estimate. □

Lemma 3.2 (Sparse-prime absence forces discrepancy; manuscript Lemma 3.2)**.** *Fix the original family and the constants of Proposition 3.1. There is a degree-dependent $c_* > 1$ such that, for $1 < c < c_*$ and sufficiently large N, the parameter hypotheses of Proposition 3.1 imply the following. If $A \subset [N]$ contains no pairwise-distinct configuration*

$$x,\ x + P_{1,Q}(y), \dots, x + P_{\ell,Q}(y), \quad y \geq 1, \quad Qy + 1 \in \mathbb{P}_c,$$

then its prime-weighted discrepancy satisfies the hypothesis of Proposition 3.1.

Status: open; collision and higher-prime-power estimates checked, discrepancy assembly unfinished.

Main term and the three errors. Choose small fixed $\tau > 0$. For $y \leq \lfloor \tau M \rfloor$, all shifts have absolute value at most $N/10$, so at least $N/2$ base points survive. The ordinary-prime model count is therefore $\gg_{\mathbf{P}} \delta^{\ell+1} \mathbb{E}_{y \leq M} \Lambda(Qy + 1)$, by the prime-mass truncation comparison.

Comparison changes the count in A to the Λ_γ count with error $O(N^{-\xi})$. Its noncollision prime terms vanish by hypothesis. Collisions occur at $O_{d,\ell}(1)$ roots of nonzero difference polynomials, contributing

$$\ll (QM)^{1-\gamma} \log(2QM)/M.$$

Higher prime powers contribute

$$\ll Q^{3/2-\gamma} M^{1/2-\gamma} (\log(2QM))^2.$$

All three errors are $O(N^{-\eta})$ for a positive η. The lower prime-mass bound $\mathbb{E}\Lambda(Qy + 1) \gg \exp(-O((\log N)^{2c_0}))$ and the density range make the errors negligible compared with the model. Subtracting the actual count from that model proves the discrepancy. □

Iteration proving Theorem 1.1

Fix $s = \lceil K_1 \alpha^{-C_1} \rceil$ and $Q_0 = \mathcal{L}(s)$, and start on a densest full residue-class intersection. At every increment set $Q_{j+1} = Q_j^{b_j+1}$ and pull A_j back to the selected progression. Exact nested rescaling preserves absence and distinctness. The modulus remains a power of Q_0; s remains fixed. Quantitative length recursion and $\log Q_j \leq \alpha^{-O(1)}$ show that every step is admissible when $\alpha \geq (\log N)^{-\kappa}$ and κ is small enough. After $O(\log \alpha^{-1} + 1)$ increments, the density exceeds 1. The recursion/admissibility proof is still open in Lean; the final finite-growth contradiction is only an auxiliary.

Lean obligation. Construct the finite iteration from the increment theorem and prove uniform admissibility at every stage. No assumption of this paper's recurrence theorem, nor an equivalent iteration-completion hypothesis, is permitted.

4 Pointwise comparison and unweighting

Definition D.4 (Orbit products and averages)**.** After choosing bounded representatives and scaling, suppose $|f_i| \leq 1$. Define

$$z_n(x) = \prod_{i=1}^{\ell} f_i(T^{P_i(n-1)}x), \quad B_N^\gamma = N^{-1} \sum_{n=1}^{N} \Lambda_\gamma(n) z_n, \quad B_N = N^{-1} \sum_{n=1}^{N} \Lambda(n) z_n.$$

Then $|z_n| \leq 1$.

The zero phase and PNT give

$$N^{-1}\sum_{n\le N}\Lambda_\gamma(n)=N^{-1}\sum_{n\le N}\Lambda(n)+O(N^{-\sigma_d})\longrightarrow 1.$$

Proposition 4.1 (Weighted pointwise comparison; manuscript Proposition 4.1)**.** *After decreasing $c_*>1$, for every $1<c<c_*$ one has $B_N^\gamma-B_N\to 0$ almost everywhere.*

Status: checked conditional proof from SS, OR and PNT.

Power saving, subsequence and interpolation. Apply the orbit estimate with $g=D_\gamma$ and

$$B=C_cN^{1-\gamma}\log(2N).$$

Theorem 2.1 then gives

$$\|B_N^\gamma-B_N\|_1\ll\log(2N)\left[N^{1-\gamma-1/K}+N^{(1-\gamma)(1-1/K)-\sigma_d/K}\right]\ll N^{-\eta}.$$

It suffices that $1-\gamma\le\min(1,\sigma_d)/(2K)$; one can take $\eta=\min(1,\sigma_d)/(4K)$ after absorbing the logarithm. Choose $A>\max(1,\eta^{-1})$, $N_j=\lfloor j^A\rfloor$. The L^1 norms are summable on this subsequence, so Tonelli gives a.e. convergence to zero there. For either nonnegative weight w, if $N_j\le N<N_{j+1}$,

$$|B_N^w-B_{N_j}^w|\le\frac{N_{j+1}-N_j}{N_j^2}\sum_{n\le N_j}w(n)+\frac{1}{N_j}\sum_{N_j<n\le N_{j+1}}w(n)=o(1).$$

The last step uses only the full mass asymptotic and $N_{j+1}/N_j\to 1$. Apply it to both weights. □

From weighted to prime-counting averages

By MW the ordinary weighted averages B_N converge a.e.; thus Proposition 4.1 gives a common limit $L(x)$ for B_N^γ. Higher prime powers satisfy

$$\sum_{\substack{p^k\le N\\k\ge 2}}\left(\Lambda_\gamma(p^k)+\Lambda(p^k)\right)\ll_\gamma N^{3/2-\gamma}(\log(2N))^2=o(N)$$

when $1/2<\gamma\le 1$. The checked cardinality bound is $\sqrt{N}\,\log_2 N$, and the combined mass bound has coefficient $(\gamma^{-1}+1)/\log 2$. The actual normalized mass tends to zero in `PrimePowers.lean`. Removing it leaves primes exactly. Center at $L(x)$ and use $a(t)=\gamma^{-1}t^{1-\gamma}\log t$. Then $W(t)=\sum_{p\le t,p\in\mathbb{P}_c}a(p)\sim t$ and the centered weighted sum is $o_x(W(t))$. Summation by parts, $a'>0$, and the same identity for W bound the unweighted centered sum by $C_{x,\varepsilon}+\varepsilon\pi_c(N)$. The mass asymptotic forces $\pi_c(N)\to\infty$, so division gives convergence to $L(x)$. The ordinary-prime calculation uses $a(t)=\log t$. The actual mass, prime-power residual, a.e. assembly and cutoff unweighting applications are checked.

Checked assembly of Theorem 1.2. `pointwiseErgodicTheorem_of_external_inputs` combines the preceding arithmetic, interpolation and unweighting steps and the bounded-representative transfer. Its only substantial external premises are SS, OR, PNT and MW. The source-to-interface bridges OR-B and MW-B remain external and are catalogued separately. □

External dependencies, correspondence and remaining work

External input E.1 (KMTT comparison and weak regularity)**.** KM-U and KM-R are separate external finite estimates with the parameter ranges recorded in Proposition 3.1 and the shared catalogue. They do not assume a PS-prime configuration or the conclusion of Theorem 1.1.

External input E.2 (Orbit estimate)**.** The earlier orbit estimate OR bounds the L^1 norm by the scaled TV expression for any weight with the specified size and phase correlations.

External input E.3 (Ordinary weighted pointwise convergence)**.** MW gives a.e. convergence of the ordinary von Mangoldt-weighted averages under the finite-L^q hypotheses; the probability-space bounded-input specialization is a bridge.

External input E.4 (Prime number theorem)**.** $N^{-1}\sum_{n\le N}\Lambda(n) \to 1$. A pinned-Mathlib theorem can replace this external entry only after its exact normalization and hypotheses are checked.

Shared manuscript correspondence

This table is identical in the two blueprints. A specification is a proposition-valued definition, not a theorem establishing that proposition. A quoted mathematical input is not a Lean axiom. The declaration catalogue and actual build evidence in the repository determine the checked status of auxiliary declarations.

Paper	Source label	Implementation boundary
Thm. 1.1	`t:szemeredi`	`PSPrimes.QuantitativeRecurrence`: faithful specification; theorem proof open.
Thm. 1.2	`t:pointwise`	`pointwiseErgodicTheorem_of_external_inputs`: checked full uniform specification from SS, OR, PNT and MW.
Thm. 1.3	`t:teravainen`	External finite generalized von Neumann estimate (TV); no project proof.
Thm. 2.1	`t:shao-sun`	External directly quoted polynomial-phase input SS; integer and small-scale translation included externally.
Lemma 2.2	`l:progression-phase`	Checked conditional theorem `progressionPhase_of_shaoSun`; no additional arithmetic input beyond SS specialization.
Prop. 2.3	`p:comparison`	`ps_counting_comparison`: exact original finite estimate checked conditionally on SS and TV; TV-B external.
Prop. 3.1	`p:KMTT`	Composite deduction from two distinct KMTT inputs; local-factor/density algebra auxiliary, complete proposition open.
Lemma 3.2	`l:PS-discrepancy`	Original discrepancy deduction; collision and higher-prime-power estimates checked; discrepancy assembly open.
Prop. 4.1	`p:pointwise-comparison`	`ps_orbit_comparison`: exact original comparison checked from SS, OR and PNT.

The nine numbered statements above are all the numbered theorem-like environments in the manuscript. The numbering is taken from the actual compiled source, including its global equation numbers (1)–(39). The auxiliary names mentioned here do not claim completion of any manuscript statement.

Shared external-input catalogue

The five central interfaces are implemented and uninstantiated.

Their source files are `ExternalInputs.lean` and `OrbitInputs.lean`. KM-U, KM-R and KM-M remain planned interfaces. A catalogue entry is not a Lean axiom; the exact actual types and dependency information are in `docs/ASSUMPTIONS.md`.

SS is the directly published polynomial-phase consequence of Shao–Sun, arXiv:2605.18676v1, Theorem 2.1 and Definition 4.5. **TV** is Teräväinen, arXiv:2401.03174v2, Theorem 4.1. The universal-phase/supremum bridge TV-B remains external. SS and TV give Proposition 2.3; the pointwise theorem does not require TV.

KM-U is KMTT, arXiv:2608.19525v1, Proposition 5.2; **KM-R** is Lemma 5.3 with its small-scale alternative and coordinate-dependent resolutions; compatibility uses (5.35)–(5.36). **KM-M** records (5.2) and (5.15). These interfaces and the original local-factor deduction and quantitative iteration remain open. Only recurrence uses KM.

OR is Mousavi, arXiv:2609.01847v1, Proposition 1.4, via `OrbitVNInput`. **MW** is Mirek–Wan–Wright, arXiv:2505.15549v2, Theorem 1.1(ii), via `OrdinaryPrimePointwiseInput`. Their source translations OR-B and MW-B remain external. **PNT** is the ordinary von Mangoldt mass limit, directly stated in Tao's 246B Notes 4, Theorem 6. Theorem 1.2 has exactly SS, OR, PNT and MW as its substantial external premises. The optional no-Siegel footnote is not used.

Primary sources were read in the cited versions. Provenance verification does not formalize an external theorem or discharge its source-formulation bridges. See the repository assumption catalogue and reference review for the actual interfaces.

What this checkpoint does and does not establish

The four conditional proofs are Theorem 1.2, Lemma 2.2 and Propositions 2.3 and 4.1. The five external interfaces are uninstantiated, with formulation bridges catalogued separately. Only Theorem 1.1, Proposition 3.1 and Lemma 3.2 remain open among the original arguments. Theorems 1.3 and 2.1 are quoted externally.